

BY SIR OTONDE MICHAEL-DELEGATE

SZL 210 PAST PAPERS WITH ANSWERS.

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SECTION A

1. State five characteristics of a typical foraminiferan. (5 marks)

1. Possess a shell (test) made of calcium carbonate or agglutinated particles.
2. Tests are usually chambered and may be perforated with small pores (foramina).
3. Move and feed using thread-like pseudopodia (reticulopodia).
4. Mostly marine, occurring as benthic or planktonic forms.
5. Life cycle often involves alternation of sexual and asexual generations.

2. Describe sexual reproduction in sessile marine invertebrates. (5 marks)

- Most sessile marine invertebrates are unable to move to find mates and therefore release gametes into the surrounding water (broadcast spawning).
- Fertilization is external.
- Reproductive activities are often synchronized by environmental cues such as temperature, tides, or lunar cycles.
- Many species are hermaphroditic, increasing the chances of successful fertilization.
- The resulting larvae are usually planktonic before settling and metamorphosing into sessile adults.

3. Explain the filter-feeding mechanisms of sessile invertebrates living in stagnant water. (5 marks)

- In stagnant water, food particles tend to settle, so filter feeders generate their own water currents.
- Water movement is produced using cilia, flagella, or muscular actions.
- Examples include choanocytes in sponges, ciliated gills in bivalves, and cirri in barnacles.

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- Food particles are trapped by mucus or ciliary action and transported to the mouth.
- This mechanism allows efficient feeding even in low-water-flow conditions.

4. (a) Describe the type of symmetry found in adult cnidarians.

(b) State two advantages of this type of symmetry. (5 marks)

(a) Adult cnidarians exhibit **radial symmetry**, with body parts arranged around a central (oral–aboral) axis.

(b) Advantages:

1. Enables detection and response to stimuli from all directions.
2. Facilitates prey capture from any direction, especially in sessile or slow-moving organisms.

5. Explain the adaptation of marine invertebrates to a benthic mode of life. (5 marks)

- Presence of attachment structures such as holdfasts, byssal threads, and adhesive discs.
- Burrowing adaptations including streamlined bodies, strong musculature, and mucus secretion.
- Protective features such as shells, spines, or camouflage to reduce predation.
- Specialized feeding methods such as deposit feeding, filter feeding, and scavenging.
- Ability to tolerate variations in temperature, salinity, and oxygen availability.

6. Describe dimorphism in cnidarians. (5 marks)

- Dimorphism refers to the occurrence of two distinct body forms in the life cycle: **polyp** and **medusa**.
- The polyp is sessile, cylindrical, and reproduces asexually, usually by budding.
- The medusa is free-swimming, bell-shaped, and reproduces sexually.

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- In some groups such as Hydrozoa, both forms occur, while in others like Anthozoa only the polyp form exists.
- Dimorphism allows exploitation of different ecological niches.

7. Explain the mode of excretion and osmoregulation in freshwater and marine protozoans. (5 marks)

- **Freshwater protozoans:** They are hyperosmotic to their environment, so water enters the cell by osmosis. Excess water is removed by contractile vacuoles to maintain osmotic balance.
- **Marine protozoans:** They are isosmotic or slightly hypoosmotic to their environment; therefore, contractile vacuoles are absent or less active.
- In both freshwater and marine protozoans, nitrogenous waste (mainly ammonia) is excreted by diffusion through the cell membrane.

8. Outline the characteristic features of a typical deuterostome. (5 marks)

1. The blastopore develops into the anus, while the mouth forms secondarily.
2. Cleavage is radial and indeterminate.
3. The coelom is formed by enterocoely.
4. Members include the phyla Echinodermata, Chordata, and Hemichordata.
5. Embryonic development is regulative rather than mosaic.

SECTION B

9. Write an essay on Poriferans. (30 marks)

Introduction

Poriferans, commonly known as sponges, belong to the phylum **Porifera** and represent one of the most primitive groups of multicellular animals. They are aquatic, mostly marine organisms that are sessile and adapted for filter feeding. Poriferans lack true tissues and organs, making them important for understanding early metazoan evolution.

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General Characteristics

- Multicellular organisms without true tissues or organs.
- Mostly asymmetrical, though some exhibit radial symmetry.
- Sessile adults attached to substrates.
- Aquatic in habitat, predominantly marine with few freshwater species.
- Body supported by spicules (calcium carbonate or silica) and/or spongin fibres.

Body Structure and Cellular Organization

The sponge body is organized around a **canal system** adapted for filter feeding and consists of:

- **Pinacoderm:** Outer layer of pinacocytes for protection.
- **Choanoderm:** Inner layer of choanocytes whose flagella generate water currents and trap food.
- **Mesohyl:** Gelatinous matrix containing amoebocytes, sclerocytes, and spongocytes.

Types of canal systems:

- **Asconoid:** Simplest, with a single spongocoel.
- **Syconoid:** Body wall folded into radial canals.
- **Leuconoid:** Highly complex, with many chambers and canals.

Nutrition and Digestion

Poriferans are filter feeders:

- Water enters through **ostia**.
- Choanocytes trap food particles.
- Digestion is **intracellular**.
- Nutrients are distributed by amoebocytes.
- Water and waste exit through the **osculum**.

Reproduction

Asexual reproduction occurs by:

- Budding

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- Fragmentation
- Gemmule formation (freshwater sponges)

Sexual reproduction:

- Mostly hermaphroditic.
- Sperm released into water; eggs retained.
- Fertilization is internal.
- Larvae (parenchymula or amphiblastula) are free-swimming before settling.

Classification

1. **Calcarea:** Calcareous spicules; marine.
2. **Hexactinellida:** Siliceous, six-rayed spicules; deep sea.
3. **Demospongiae:** Siliceous spicules and/or spongin; largest group.
4. **Homoscleromorpha:** Small marine group with simple organization.

Ecological and Economic Importance

- Filter water and improve water quality.
- Provide habitats for small organisms.
- Some bore into limestone and coral.
- Source of spongin and bioactive compounds used in medicine.

Conclusion

Poriferans, though structurally simple, are ecologically significant and evolutionarily important. Their body organization and feeding strategy illustrate early adaptations to multicellular life.

10. Describe the common and distinguishing features between insects and spiders. (30 marks)

Introduction

Insects (Class Insecta) and spiders (Class Arachnida) are members of the phylum Arthropoda. While they share basic arthropod features, they differ markedly in body organization, physiology, and lifestyle.

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Common Features

- Chitinous exoskeleton.
- Segmented body.
- Jointed appendages.
- Open circulatory system.
- Ventral nerve cord.
- Bilateral symmetry.
- Growth by moulting (ecdysis).

Distinguishing Features

Feature	Insects	Spiders
Body regions	Head, thorax, abdomen	Cephalothorax and abdomen
Legs	6 (3 pairs)	8 (4 pairs)
Antennae	Present	Absent
Wings	Usually present	Absent
Eyes	Compound and ocelli	Simple eyes
Mouthparts	Mandibles, maxillae, etc.	Chelicerae and pedipalps
Respiration	Tracheal system	Book lungs/tracheae
Feeding	Diverse diets	Predatory
Development	Metamorphosis common	Direct development

Ecological Roles

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- **Insects:** Pollination, decomposition, pests, food source.
- **Spiders:** Control insect populations as predators.

Conclusion

Although insects and spiders share a common arthropod ancestry, they exhibit distinct adaptations reflecting different ecological roles and evolutionary paths.

11. Discuss the adaptive features of echinoderms. (30 marks)

Introduction

Echinoderms are exclusively marine animals belonging to the phylum **Echinodermata**. They include starfish, sea urchins, brittle stars, sea cucumbers, and crinoids. Their unique adaptations enable survival in diverse marine habitats.

Pentaradial Symmetry

- Adult echinoderms exhibit five-part radial symmetry.
- Allows interaction with the environment from all directions.
- Suited to slow-moving or sessile lifestyles.

Endoskeleton

- Composed of calcareous ossicles.
- Provides support and protection.
- Bears spines for defence and locomotion.

Water Vascular System

- Hydraulic system used for:
 - Locomotion
 - Feeding
 - Respiration
 - Sensory functions
- Operates through tube feet powered by water pressure.

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Regeneration and Autotomy

- Ability to regenerate lost body parts.
- Some species shed limbs to escape predators.
- Enables asexual reproduction in some cases.

Feeding Adaptations

- Starfish evert stomachs to digest prey.
- Sea urchins use Aristotle's lantern for grazing.
- Sea cucumbers feed on detritus using tentacles.
- Crinoids filter plankton using arms.

Respiration and Excretion

- Gas exchange via tube feet, papulae, or respiratory trees.
- Nitrogenous waste excreted by diffusion.

Defensive Adaptations

- Spines and pedicellariae.
- Evisceration in sea cucumbers.
- Camouflage and nocturnal behavior.

Conclusion

Echinoderms display remarkable adaptations such as the water vascular system, regeneration, and specialized feeding mechanisms. These features have enabled them to occupy a wide range of marine environments successfully.

ANSWER ALL QUESTIONS (40 MARKS)

1. Outline five characteristics of chordates.

1. Presence of a **notochord** at least at some stage of development.
2. Presence of a **dorsal, hollow nerve cord**.
3. Presence of **pharyngeal gill slits** at some stage of the life cycle.
4. Presence of a **post-anal tail** extending beyond the anus.
5. Presence of an **endostyle or thyroid gland** involved in metabolism.

2. Make a clear, well-labelled diagram of *Amphioxus*.

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A labelled diagram of Amphioxus (Branchiostoma) should show the following parts:

- Rostrum
- Mouth with oral cirri
- Pharyngeal gill slits
- Atrium
- Atrio-pore
- Notochord
- Dorsal hollow nerve cord
- Myotomes (muscle blocks)
- Post-anal tail
- Anus

(In the examination, draw a neat lateral view and label all parts clearly.)

3. Sharks are well equipped for their predatory life. Justify this statement.

- **Streamlined body** reduces resistance during swimming.
- **Powerful jaws with sharp, replaceable teeth** enable efficient capture and tearing of prey.
- **Well-developed sense organs**, including keen vision and olfactory sense.
- **Lateral line system** detects vibrations and movement in water.
- **Ampullae of Lorenzini** detect weak electrical signals from prey.
- Strong tail and fins provide speed and maneuverability.

4. Explain the differences between venomous and non-venomous snakes.

Feature	Venomous Snakes	Non-Venomous Snakes
Fangs	Long, hollow or grooved fangs	No specialized fangs
Venom glands	Present	Absent
Method of killing prey	Inject venom	Constriction or swallowing whole
Head shape	Usually triangular	Usually narrow and elongated
Bite effect	Causes poisoning	Generally harmless

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5. Outline the characteristics of the hagfish.

1. Elongated, eel-like body without jaws.
2. Cartilaginous skeleton with persistent notochord.
3. Absence of paired fins and vertebrae.
4. Presence of slime glands that secrete mucus for defence.
5. Single nostril opening into the pharynx.
6. Scavenging or parasitic mode of feeding.

6. Describe briefly the adaptations of terrestrial amphibians.

- **Lungs** for breathing air.
- **Moist skin** for cutaneous respiration.
- **Strong limbs** for locomotion on land.
- **Well-developed eyes with eyelids** to prevent drying.
- **Tympanic membrane** for hearing airborne sounds.
- Ability to return to water for reproduction.

7. Describe the ecological importance of birds to man.

- **Pollination** of flowering plants.
- **Seed dispersal**, aiding plant propagation.
- **Control of insect pests** harmful to crops.
- Serve as **sources of food** (meat and eggs).
- Used in **scientific research and education**.
- Contribute to **ecotourism** and ecosystem balance.

8. Outline the distinctive features of *Archaeopteryx lithographica*.

1. Possessed **feathers** like modern birds.
2. Had **teeth in the jaws**, a reptilian feature.
3. Presence of a **long bony tail with vertebrae**.
4. Forelimbs modified into **wings with claws**.
5. Skull showed reptilian and avian characteristics.
6. Considered a **connecting link between reptiles and birds**.

9. Give a detailed account of the digestive, circulatory and reproductive systems of *Ascidia*. (30 marks)

Introduction

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Ascidia (sea squirt) is a marine, sessile tunicate belonging to the subphylum **Urochordata (Tunicata)**. Although simple in adult form, it exhibits chordate features during its larval stage. Its internal organization is well adapted to a sedentary, filter-feeding lifestyle.

Digestive System

The digestive system of *Ascidia* is complete and adapted for filter feeding.

- Food particles are drawn into the body through the **incurrent (oral) siphon**.
- The **buccal cavity** leads into a large **pharynx** perforated by numerous **gill slits (stigmata)**.
- The pharynx is lined with the **endostyle**, which secretes mucus that traps food particles.
- Ciliary action moves mucus-bound food into the **dorsal lamina**, which directs it to the **oesophagus**.
- The oesophagus opens into the **stomach**, where digestion begins.
- Further digestion and absorption occur in the **intestine**.
- Undigested materials are expelled through the **anus**, which opens near the **excurrent (atrial) siphon**.

Circulatory System

The circulatory system of *Ascidia* is simple and **open**.

- It consists of a **heart**, blood vessels, and blood sinuses.
- The heart is a **tubular structure** located ventrally near the stomach.
- A unique feature is the **reversal of blood flow**: the heart periodically changes the direction of blood pumping.
- Blood is colorless and lacks respiratory pigments in most species.
- Blood vessels distribute nutrients and oxygen to tissues and collect metabolic wastes.
- Gas exchange occurs mainly across the **pharyngeal wall** and body surface.

Reproductive System

Ascidia reproduces both **sexually and asexually**, though sexual reproduction is more common.

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Sexual Reproduction

- Most ascidians are **hermaphrodites**.
- The **gonads** consist of an ovary and a testis embedded in the body wall or mesentery.
- Sperm and eggs are released into the atrial cavity and discharged through the excurrent siphon.
- Fertilization is usually **external**.
- Development includes a free-swimming **tadpole larva** with chordate features such as:
 - Notochord
 - Dorsal hollow nerve cord
 - Post-anal tail
- The larva undergoes **retrogressive metamorphosis**, becoming a sessile adult.

Asexual Reproduction

- Occurs in colonial ascidians by **budding**.

Conclusion

The digestive, circulatory, and reproductive systems of *Ascidia* are highly specialized for a sessile, filter-feeding existence. Despite their simple adult structure, ascidians retain chordate features during larval life, highlighting their evolutionary significance.

10. Write an essay on adaptive radiation in reptiles. (30 marks)

Introduction

Adaptive radiation refers to the evolutionary diversification of a single ancestral group into various forms adapted to different environments and ecological niches. Reptiles exhibit extensive adaptive radiation, enabling them to inhabit terrestrial, aquatic, arboreal, and aerial environments.

Forms of Adaptive Radiation in Reptiles

1. Terrestrial Adaptations

- Strong, clawed limbs for walking and running.

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- Dry, keratinized skin with scales to prevent water loss.
- Efficient lungs for respiration.
- Example: Lizards, tortoises.

2. Aquatic Adaptations

- Streamlined body for swimming.
- Limbs modified into paddles or flippers.
- Presence of tail for propulsion.
- Ability to close nostrils while submerged.
- Example: Sea snakes, turtles, crocodiles.

3. Arboreal Adaptations

- Long limbs and sharp claws for climbing.
- Prehensile tails in some species.
- Good vision and balance.
- Example: Chameleons, tree lizards.

4. Fossorial Adaptations

- Elongated cylindrical body.
- Reduced or absent limbs.
- Strong skull for burrowing.
- Example: Blind snakes.

5. Aerial Adaptations

- Patagium (gliding membrane) between limbs.
- Lightweight body.
- Example: Flying lizards (*Draco*).

6. Desert Adaptations

- Nocturnal habits.
- Ability to conserve water.
- Thick, scaly skin.
- Example: Desert lizards.

Evolutionary Significance

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- Enabled reptiles to dominate land during the Mesozoic era.
- Reduced dependence on water for reproduction through shelled eggs.
- Laid evolutionary foundation for birds and mammals.

Conclusion

Adaptive radiation in reptiles demonstrates remarkable evolutionary plasticity. Through structural, physiological, and behavioral modifications, reptiles have successfully occupied nearly every terrestrial and semi-aquatic habitat on Earth.

11. Describe the general characteristics of mammals and discuss their adaptation to terrestrial life. (30 marks)

Introduction

Mammals are warm-blooded vertebrates belonging to the class **Mammalia**. They are among the most highly evolved animals, exhibiting complex structural, physiological, and behavioral adaptations that enable them to thrive in terrestrial environments.

General Characteristics of Mammals

- Body covered with **hair or fur**.
- Presence of **mammary glands** for milk production.
- **Endothermic** (warm-blooded).
- Four-chambered heart.
- Diaphragm separating thoracic and abdominal cavities.
- External ear pinnae.
- Heterodont dentition.
- Mostly viviparous with internal fertilization.

Adaptations to Terrestrial Life

1. Locomotion

- Strong limbs positioned under the body.
- Modified limbs for running, jumping, digging, or climbing.

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2. Respiration

- Well-developed lungs.
- Presence of a **diaphragm** for efficient breathing.

3. Circulation

- Four-chambered heart ensures efficient oxygen supply.

4. Temperature Regulation

- Hair and subcutaneous fat for insulation.
- Sweat and sebaceous glands for thermoregulation.

5. Excretion

- Efficient kidneys with loop of Henle for water conservation.

6. Reproduction

- Internal fertilization and placental development.
- Parental care enhances survival of young.

7. Sensory and Nervous System

- Highly developed brain and sense organs.
- Advanced learning and behavioral responses.

Conclusion

Mammals possess a combination of anatomical, physiological, and behavioral adaptations that make them highly successful terrestrial animals. Their ability to regulate body temperature, care for young, and efficiently utilize resources has allowed them to colonize diverse habitats worldwide.

1. Make a clear well-labelled diagram of *Amphioxus*. (5 marks)

*A neat lateral diagram of **Amphioxus (Branchiostoma)** should be drawn and labelled with the following parts:*

- Rostrum

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- Oral hood with cirri
- Mouth
- Pharyngeal gill slits
- Atrium
- Atrio-pore
- Notochord
- Dorsal hollow nerve cord
- Myotomes (muscle blocks)
- Post-anal tail
- Anus

(In the examination, a well-proportioned drawing with at least 8 correct labels earns full marks.)

2. Explain the following terms:

(a) Homodont teeth. (1 mark)

Homodont teeth are teeth that are **all similar in shape and size**, as seen in fishes and reptiles.

(b) Retrogressive metamorphosis. (1 mark)

Retrogressive metamorphosis is a type of development in which a **larva with complex structures transforms into a simpler adult**, as seen in *Ascidia*.

(c) Pineal eye. (1 mark)

The pineal eye is a **light-sensitive organ** found in some reptiles and fishes, associated with regulation of biological rhythms.

(d) Spiral valve. (1 mark)

The spiral valve is a **spirally folded structure in the intestine** of some fishes that increases the surface area for digestion and absorption

(e) Heterocercal tail. (1 mark)

A heterocercal tail is an **asymmetrical tail** in which the vertebral column extends into the upper lobe, common in sharks.

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3. Distinguish between hagfish and lamprey. (5 marks)

Feature	Hagfish	Lamprey
Jaws	Absent	Absent
Vertebrae	Absent	Rudimentary vertebrae present
Eyes	Poorly developed	Well developed
Mouth	Tentacles around mouth	Suctorial mouth with teeth
Lifestyle	Scavenger	Mostly parasitic

4. Explain the difference between venomous and non-venomous snakes. (5 marks)

- **Venomous snakes** possess venom glands and specialized fangs for injecting venom into prey.
- **Non-venomous snakes** lack venom glands and fangs.
- Venomous snakes kill prey by poisoning, while non-venomous snakes kill by constriction or swallowing whole.
- Venomous snakes often have triangular heads; non-venomous snakes usually have narrow heads.
- Bites of venomous snakes are dangerous, while those of non-venomous snakes are generally harmless.

5. Outline any five differences between vertebrates and invertebrate chordates. (5 marks)

Feature	Vertebrates	Invertebrate chordates
Vertebral column	Present	Absent
Notochord	Replaced by vertebrae	Persists throughout life
Brain	Well developed	Poorly developed
Skeleton	Endoskeleton present	No true endoskeleton
Heart	Well developed	Simple or absent

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6. Explain the functions of the extra-embryonic membranes in reptiles. (5 marks)

- **Amnion:** Protects the embryo and prevents desiccation.
- **Chorion:** Aids in gaseous exchange.
- **Allantois:** Stores nitrogenous waste and assists respiration.
- **Yolk sac:** Provides nourishment to the embryo.
- Together, these membranes enable reptiles to reproduce successfully on land.

7. Outline the characteristics of the existing cartilaginous fishes. (5 marks)

1. Skeleton made entirely of cartilage.
2. Body covered with placoid scales.
3. Presence of heterocercal tail.
4. Five to seven gill slits without operculum.
5. Internal fertilization common.

8. *Archaeopteryx lithographica* is believed to have originated from reptiles. Explain this statement. (5 marks)

- Presence of **teeth in the jaws**, a reptilian feature.
- Possession of a **long bony tail** with vertebrae.
- Forelimbs had **claws on the wings**.
- Skull resembled that of reptiles.
- Combined reptilian and avian features show it as a **connecting link between reptiles and birds**.

1. Discuss the traditional divisions and characteristics of the phylum Chordata. (30 marks)

Traditional divisions of Phylum Chordata

Phylum Chordata is traditionally divided into three subphyla:

(a) Subphylum Urochordata (Tunicata)

- Mostly marine animals
- Body covered by a tunic made of tunicin
- Notochord present only in the larval tail
- Adult is usually sessile

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- Examples: *Ascidia*, *Salpa*, *Doliolum*

(b) Subphylum Cephalochordata

- Small, marine, fish-like animals
- Notochord extends from head to tail throughout life
- No true head or vertebral column
- Closed circulatory system but no true heart
- Example: *Branchiostoma* (Amphioxus)

(c) Subphylum Vertebrata (Craniata)

- Notochord replaced by vertebral column in adults
- Well-developed head with brain enclosed in a cranium
- Paired appendages usually present
- Advanced organ systems
- Includes classes such as Pisces, Amphibia, Reptilia, Aves, and Mammalia

General characteristics of Phylum Chordata

1. Presence of a notochord at least at some stage of development
2. Dorsal, hollow tubular nerve cord
3. Pharyngeal gill slits present at some stage
4. Post-anal tail present at some stage
5. Bilaterally symmetrical body
6. Triploblastic organization
7. True coelom present
8. Organ-system level of organization
9. Closed circulatory system (except some lower forms)
10. Endoskeleton present (cartilaginous or bony)
11. Respiration through gills, lungs, or skin
12. Excretion mainly by kidneys
13. Sexes usually separate
14. Fertilization may be external or internal
15. Well-developed sense organs in higher forms

2. Discuss the adaptation of amphibians to terrestrial life. (30 marks)

Amphibians show several structural, physiological, and reproductive adaptations that enable them to live on land, though they still depend partly on water.

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(a) Locomotory adaptations

- Development of pentadactyl limbs for movement on land
- Strong pectoral and pelvic girdles to support body weight
- Vertebral column strengthened to bear body weight

(b) Respiratory adaptations

- Development of lungs for aerial respiration
- Moist skin allows cutaneous respiration
- Buccopharyngeal respiration supplements lung breathing

(c) Circulatory adaptations

- Three-chambered heart (two atria and one ventricle)
- Partial double circulation improves oxygen supply
- Separation of oxygenated and deoxygenated blood

(d) Excretory adaptations

- Kidneys adapted to excrete urea, reducing water loss
- Urinary bladder stores urine temporarily

(e) Sensory adaptations

- Well-developed eyes with eyelids and nictitating membrane
- Tympanic membrane for hearing airborne sounds
- Olfactory organs adapted for air

(f) Skin adaptations

- Skin has mucus glands to prevent desiccation
- Skin plays a role in respiration

(g) Reproductive adaptations (limited terrestrial adaptation)

- Eggs lack shells and must be laid in water
- External fertilization common
- Larval stage (tadpole) aquatic, adult semi-terrestrial

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3. Give a detailed account of the digestive, circulatory and reproductive systems of *Ascidia*. (30 marks)

(a) Digestive system of *Ascidia*

- Incomplete but well-developed digestive tract
- Mouth opens into a large pharynx (branchial sac)
- Pharynx perforated by numerous gill slits
- Food particles trapped in mucus secreted by endostyle
- Cilia move food to dorsal lamina
- Food passes into oesophagus, stomach, intestine
- Undigested waste expelled through anus into atrial cavity

(b) Circulatory system of *Ascidia*

- Open type of circulatory system
- Heart is tubular and lies ventral to the pharynx
- Blood vessels poorly developed
- Heart shows periodic reversal of blood flow
- Blood contains colourless cells; respiratory pigments usually absent
- Circulation helps in transport of nutrients and gases

c) Reproductive system of *Ascidia*

- Hermaphroditic; both testes and ovary present in same individual
- Gonads lie near intestinal loop
- Fertilization usually external
- Eggs released into water through atrial siphon
- Development includes a free-swimming tadpole larva
- Larva shows chordate characters such as notochord, dorsal nerve cord, and tail
- Metamorphosis leads to sessile adult form

1. Outline the characteristics of vertebrates

- Presence of a vertebral column replacing the notochord partially or completely
- Body divided into head, trunk, and usually tail
- Well-developed brain enclosed in a cranium
- Dorsal hollow nerve cord forming brain and spinal cord
- Endoskeleton made of cartilage or bone

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- Paired appendages (fins or limbs) usually present
- Closed circulatory system with a ventral heart
- Blood contains red blood cells with haemoglobin
- Respiration through gills or lungs
- Excretion mainly by kidneys
- Sexes usually separate
- Well-developed sense organs

2. Make a clear, well labelled diagram of a named cephalochordate

Named cephalochordate: *Branchiostoma* (Amphioxus)

(Textual description for examination use)

A well-labelled diagram should show:

- Rostrum
- Oral hood
- Buccal cirri
- Pharynx with gill slits
- Notochord extending entire length
- Dorsal hollow nerve cord
- Myotomes
- Atrium
- Atriopore
- Hepatic caecum
- Intestine
- Anus
- Post-anal tail

3. Outline any five features that distinguish mammals from other vertebrates

- Presence of mammary glands for milk production
- Body covered with hair or fur
- Presence of a muscular diaphragm separating thorax and abdomen
- Three middle ear ossicles (malleus, incus, stapes)
- Endothermic (warm-blooded) with constant body temperature

4. Outline the differences between Chondrichthyes and Osteichthyes

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Feature	Chondrichthyes	Osteichthyes
Endoskeleton	Cartilaginous	Bony
Scales	Placoid scales	Cycloid, ctenoid, or ganoid
Gill openings	5–7 gill slits, no operculum	Gills covered by operculum
Swim bladder	Absent	Present
Fertilization	Mostly internal	Mostly external

5. Sharks are successful predators. Justify this statement

- Streamlined body reduces water resistance
- Powerful muscular tail for fast swimming
- Sharp, replaceable teeth for gripping and cutting prey
- Highly developed sense organs, including lateral line system
- Acute sense of smell and electroreception (ampullae of Lorenzini)
- Strong jaws and aggressive feeding behavior

6. Describe briefly the adaptations of terrestrial amphibians

- Development of limbs with digits for movement on land
- Lungs for aerial respiration
- Moist skin allowing cutaneous respiration
- Strong pectoral and pelvic girdles for body support
- Three-chambered heart for partial double circulation
- Sense organs adapted for air, including eyes and ears

7. Outline the life-cycle of a typical anuran

- Eggs laid in water
- External fertilization
- Eggs hatch into tadpole larvae
- Tadpole breathes using gills and has a tail
- Gradual metamorphosis occurs
- Development of limbs
- Tail resorbed
- Gills replaced by lungs
- Adult frog becomes semi-terrestrial

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8. Explain the economic importance of birds

- Source of food (meat and eggs)
- Control of insect pests
- Pollination and seed dispersal
- Source of manure (guano)
- Feathers used for decoration and insulation
- Birds contribute to tourism and recreation
- Used in scientific research and education

9. Write an essay on Agnathans. (30 marks)

- Agnathans are primitive vertebrates characterized by the absence of jaws. They represent the earliest group of vertebrates and provide important evidence on the origin and early evolution of vertebrates. The living agnathans are represented by lampreys and hagfishes, which are placed under the superclass Agnatha.
- Agnathans have an elongated, eel-like body without paired fins. Instead of jaws, they possess a circular, suctorial mouth adapted for feeding by attachment or scavenging. The endoskeleton is cartilaginous, and the notochord persists throughout life without being replaced by a true vertebral column. Cranium is present, enclosing a simple brain.
- The skin of agnathans is smooth, naked, and lacks scales. Numerous slime glands are present in hagfishes, producing mucus for defense. Respiration occurs through gills, which are arranged in pouch-like structures opening externally by several gill slits. There is no operculum.
- The circulatory system is closed, with a two-chambered heart consisting of one atrium and one ventricle. Blood contains haemoglobin dissolved in plasma rather than within red blood cells in some species. Excretion is carried out by kidneys, mainly mesonephric in nature.
- The nervous system is simple, with poorly developed sense organs. Eyes may be reduced or absent, especially in hagfishes. The lateral line system is present in lampreys.
- Reproduction is sexual, with sexes separate. Fertilization is external, and development is indirect. Lampreys exhibit a distinct larval stage called ammocoetes, which undergoes metamorphosis into the adult form.
- In conclusion, agnathans are important from an evolutionary perspective as they represent the most primitive living vertebrates and provide insight into the early stages of vertebrate evolution.
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- **10. Discuss the digestive, respiratory and circulatory systems of a named tunicate. (30 marks)**

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- *Named tunicate: Ascidia*

- **Digestive system**

The digestive system of *Ascidia* is complete and adapted to filter feeding. Food enters through the oral siphon into the pharynx or branchial sac, which is perforated by numerous gill slits. The endostyle, located on the ventral side of the pharynx, secretes mucus that traps food particles. Ciliary action carries the mucus-bound food to the dorsal lamina and then into the oesophagus. Food passes into the stomach, where digestion occurs, followed by absorption in the intestine. Undigested food is expelled through the anus into the atrial cavity and exits via the atrial siphon.

- **Respiratory system**

Respiration in *Ascidia* takes place mainly through the pharyngeal wall. Water enters through the oral siphon, passes through the gill slits into the atrial cavity, and leaves via the atrial siphon. Exchange of gases occurs as water passes over the richly vascularized pharyngeal walls. Oxygen diffuses into the blood, while carbon dioxide diffuses out.

- **Circulatory system**

The circulatory system is of the open type. The heart is tubular and lies ventral to the pharynx. It is enclosed in a pericardial cavity and shows periodic reversal of blood flow. Blood vessels are poorly developed, and blood flows through sinuses. The blood lacks respiratory pigments in most species, and circulation mainly functions in the transport of nutrients and gases.

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- **11. Discuss the general characteristics of reptiles and their adaptation to terrestrial life. (30 marks)**

- Reptiles are cold-blooded vertebrates that are well adapted to life on land. They were the first vertebrates to become completely independent of water for reproduction.

- **General characteristics of reptiles**

Reptiles have dry skin covered with horny epidermal scales that reduce water loss. The body is divided into head, neck, trunk, and tail. Limbs, when present, are usually pentadactyl and clawed. The endoskeleton is well ossified, and the skull has a single occipital condyle.

- Respiration occurs exclusively through lungs, which are well developed with increased internal surface area. The heart is usually three-chambered, except in crocodiles where it is four-chambered. Circulation is double but incomplete in most reptiles. Excretion is carried out by metanephric kidneys, and nitrogenous waste is excreted mainly as uric acid.

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- Reptiles are ectothermic, relying on external heat sources to regulate body temperature. Sexes are separate, and fertilization is internal. Most reptiles lay shelled eggs with extra-embryonic membranes such as amnion, chorion, and allantois.
- **Adaptations to terrestrial life**
Dry, scaly skin prevents desiccation. Well-developed lungs ensure efficient aerial respiration. Limbs and strong girdles support the body and allow efficient movement on land. Internal fertilization and shelled eggs allow reproduction away from water. Excretion of uric acid conserves water. Sense organs, especially eyes and olfactory organs, are adapted for life in air.
- In summary, reptiles exhibit structural, physiological, and reproductive adaptations that enable them to survive and reproduce successfully in terrestrial environments.



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